MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT AND ENGINEERING

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A REPORT

ON

PROPHET MODEL IMPLEMENTATION FOR DIGITAL PAYMENT VOLUME FORECASTING WITH HOLIDAY AND EVENT EFFECTS

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1.0 Abstract

Daily forecasts of digital payment volume help banks and payment networks plan server capacity, fraud monitoring and settlement liquidity. Daily volume depends strongly on the calendar: weekends, salary days, festivals and online sales all move it. This report implements the Prophet model to forecast daily Unified Payments Interface (UPI) transaction volume and examines the contribution of holiday and event information. Since the National Payments Corporation of India (NPCI) publishes UPI figures monthly, the study uses a synthetic daily series anchored to real NPCI monthly totals from January 2022 to August 2026, with holiday and event effects of known size. The original evaluation compared a seasonal-naive baseline, SARIMAX and three Prophet variants using rolling-origin cross-validation. The revised notebook extends the evaluation with WAPE, sMAPE and mean forecast error, compares performance at 7-, 30- and 90-day lead horizons, reports fold-level variability, and applies an approximate lognormal mean correction to SARIMAX forecasts on the original transaction scale. The previously reported baseline results showed Prophet with holidays and events achieving a MAPE of 2.45%, compared with 5.54% for seasonal naive and 4.96% for SARIMAX. These figures are from the original run and must be regenerated before being treated as results of the revised pipeline, because the SARIMAX correction and expanded evaluation can change the metrics. All conclusions remain limited to the synthetic data-generating assumptions and are not claims about real-world daily UPI accuracy.

2.0 Introduction

The Unified Payments Interface (UPI), launched by NPCI in 2016, has become India's main channel for digital payments. Monthly UPI volume grew from about 4.6 billion transactions in January 2022 to about 24.5 billion in August 2026, more than five times in under five years. At this scale, even a small forecasting error means millions of transactions that servers, fraud systems and settlement teams were not prepared for.

Daily payment volume is not smooth. It is higher on Saturdays and lower on Sundays, rises when salaries are credited at the start of the month, ramps up for about two weeks before Diwali, dips on national holidays and jumps during e-commerce sales. A forecast that ignores these calendar effects is wrong on exactly the days that matter most.

Prophet, developed at Facebook (now Meta), is a forecasting model built for this kind of business data. It splits a series into a trend, seasonal patterns, holiday effects and extra regressors, and it lets the analyst list holidays and events directly. This project tests how well Prophet uses that calendar information. It addresses four questions:

- Does adding holidays and events to Prophet improve forecast accuracy, and on which days?
- Does Prophet recover the true size of each holiday and event effect?
- How does Prophet compare with SARIMAX, the classical model covered in the course?
- What does the model forecast for September 2026 to February 2027?

Section 3 reviews the relevant literature. Section 4 describes the data, Section 5 the methods and evaluation metrics, Section 6 the results and additional horizon/fold analyses, Section 7 discusses the findings and limitations, and Section 8 concludes.

3.0 Literature Review

3.1 ARIMA and SARIMAX Models

The Box-Jenkins approach models a series through its own past values (autoregressive terms), past errors (moving average terms) and differencing to remove trends [2]. Seasonal ARIMA (SARIMA) adds the same terms at a seasonal lag, such as 7 days for weekly patterns. SARIMAX further adds exogenous variables, for example holiday indicators. Model orders are usually chosen by the Akaike Information Criterion (AIC) [10], and the residuals are checked with the Ljung-Box test for leftover autocorrelation [7]. These models describe short-term dependence well, but their forecast uncertainty grows quickly with the forecast horizon.

3.2 The Prophet Model

Taylor and Letham [1] proposed Prophet as a decomposable model: a piecewise-linear trend whose slope can change at points called changepoints, seasonal patterns written as Fourier series, holiday effects with a window of days around each date, and optional regressors. The model is fitted in the probabilistic language Stan, with prior distributions that keep each component from overfitting. Prophet was designed so that analysts with domain knowledge, such as which festivals matter, can feed that knowledge into the model directly instead of tuning ARIMA orders.

3.3 Holiday and Event Effects

Calendar effects are central in retail sales, electricity demand and payment volumes [3]. Festivals that follow the lunar calendar, such as Diwali, Holi and Eid, move by several weeks from year to year, so a fixed yearly seasonal pattern cannot capture them. The standard approach is to add indicator variables for each holiday, often for several days before and after it, so that build-up and recovery effects are captured [3].

3.4 Forecast Evaluation

A model that fits its training data well does not necessarily forecast well. Tashman [9] recommends rolling-origin evaluation: the model is trained up to a cutoff date, forecasts the period after it, and the cutoff is then moved forward. Every error is measured on data the model has not seen. We use this design throughout.

3.5 Research Gap

There is no public daily UPI series, so the effect of festivals on daily UPI volume has not been measured directly. More generally, with real data the true size of a holiday effect is never known, so it is hard to tell whether a model estimates it correctly. A synthetic series with known effects, anchored to real monthly totals, allows both forecast accuracy and effect recovery to be tested.

4.0 Data

4.1 NPCI Monthly Data

NPCI publishes UPI ecosystem statistics on its website [11], one month at a time. We collected total, person-to-person (P2P) and person-to-merchant (P2M) volume and value for January 2022 to August 2026 (56 months), using an automated browser (Playwright) because the page is rendered with JavaScript. Every month passed the check P2P + P2M = total. Data before 2022 sits in an archived view that could not be collected reliably, so the study starts in January 2022, after the COVID-19 lockdowns.

4.2 Synthetic Daily Series

Prophet's holiday modelling needs daily data, because a two-week festival effect disappears inside a monthly total. We therefore generated a daily series in which each day's volume is the product of six components:

$$y(t) = trend(t) \times weekday(t) \times monthday(t) \times holiday(t) \times sale(t) \times noise(t)$$

Table 1: Components of the synthetic daily series

Component	How it is built	Size
Trend	Smooth curve through NPCI monthly averages (log scale)	About 150 to 790 million a day
Weekday	Fixed multiplier per day of week	Saturday +3%, Sunday -7%
Payday	Days 1 to 5 of each month	+6%
Month-end	Last two days of each month	+3%
Holidays	Festival and national holiday windows (Table 2)	-10% to +20%
Sale	One 4 to 6 day e-commerce sale per year	+8% to +12%
Noise	Autocorrelated noise, AR(1)	phi = 0.3, sigma = 2.5%

Because every effect is injected with a known size, the true effect on each day is stored with the data. This is the ground truth that Prophet's estimates are compared against in Section 6.4.

4.3 Anchoring to NPCI Totals and a Correction

Each month of the synthetic series is scaled so that it sums exactly to the real NPCI total. In the first version of the generator, this was applied to every month, including festive months. NPCI's real data shows a festive bump of the injected size only in October 2024, so forcing the other festive months to match NPCI pushed their ordinary days about 5% below trend. Prophet then saw Diwali as only about 4% instead of the true 9.1%.

We corrected the generator. Five festive months (October 2022, October 2023, November 2023, October 2024 and October 2025) are no longer forced to match NPCI. Their ordinary days follow the trend and the festival sits on top. All other months still match NPCI exactly, and the festive months differ from it by between -1.4% and +6.5%.

4.4 Holidays and Events

Table 2 lists the calendar effects given to the models. Holidays are recurring public or cultural dates, each with a window of days around it. Events are other calendar effects: the yearly e-commerce sale, payday and month-end. Other gazetted holidays such as Dussehra and Christmas were also given to the models as single days. Their true effect is zero, which tests whether a model invents effects that are not there.

Table 2: Holidays and events modelled

Name	Type	Window (days)	True effect
Diwali	Holiday	-10 to +3	Ramp to +20%, then a dip
Holi	Holiday	-2 to +1	+8% before, -10% on the day
Eid al-Fitr	Holiday	-3 to 0	Ramp to +6%
Raksha Bandhan	Holiday	-3 to 0	Ramp to +5%
Republic Day, Independence Day, Gandhi Jayanti	Holiday	0	-5%
E-commerce sale	Event	4 to 6 days	+8% to +12%
Payday	Event	Days 1 to 5	+6%
Month-end	Event	Last 2 days	+3%

5.0 Methodology

5.1 Exploratory Analysis

The series was examined with an STL decomposition on the log scale with a weekly period [4], the Augmented Dickey-Fuller (ADF) test [5] and the KPSS test [6] for stationarity, ACF and PACF plots, and average weekday and day-of-month profiles.

5.2 Prophet

In multiplicative mode, Prophet models the series as:

$$y(t) = g(t) \times [\ 1 + s(t) + h(t) + \beta x(t)\ ] + \varepsilon(t)$$

Here g(t) is the trend, s(t) the weekly and yearly seasonality, h(t) the holiday and event effects, x(t) the payday and month-end regressors with coefficients β, and ε(t) the error. Each day of a holiday window gets its own coefficient, so Prophet can learn the shape of a festival, such as Diwali's ten-day build-up.

Multiplicative mode was used because volume grew more than five times, and a festival adds a percentage of current volume rather than a fixed number of transactions. Payday and month-end were added as regressors because they are known in advance. A smooth trend (changepoint prior scale 0.005) and a low-order yearly pattern (Fourier order 3) performed best in a preliminary search. Three versions were built: Prophet base (trend and seasonality only), Prophet + holidays, and Prophet + holidays + events.

5.3 SARIMAX

SARIMAX was fitted to log volume as a regression with seasonal ARIMA errors. Its exogenous variables carry the same calendar information as the full Prophet model: one indicator for each day of each holiday and sale window, payday and month-end indicators, and yearly Fourier terms. Following the course procedure, the order was chosen by fitting combinations of p and q from 0 to 2 and P and Q from 0 to 1, with d = 1, D = 0 and a seasonal period of 7. The lowest-AIC candidate was selected using only the first two years (2022–2023), so the evaluation period did not influence the order choice. For forecasting, SARIMAX predicts log volume. The revised notebook applies an approximate lognormal mean correction, $\exp(\mu + 0.5\sigma^2)$, using the training residual variance, before converting forecasts back to the transaction scale. This correction is an approximation and is retained only if held-out validation supports it; it does not guarantee improved accuracy.

5.4 Baseline

The seasonal naive baseline forecasts each day with the value from the same weekday in the last week of training data. Any useful model must beat it.

5.5 Evaluation Design and Metrics 7

All models were evaluated with rolling-origin cross-validation [9]. Each model was trained up to a cutoff and forecast the following 90 days; the cutoff then advanced by 90 days. The initial training period was two years, and all models used the same forecast origins. In addition to the primary 90-day evaluation, the revised notebook summarizes pooled errors for the first 7, 30 and 90 lead days, allowing short-, medium- and longer-horizon performance to be compared. Fold-level MAPE,

WAPE, MAE and RMSE are also summarized using the mean, standard deviation, minimum and maximum across folds. Because forecast windows overlap, these fold summaries are descriptive measures of variability and are not treated as independent-sample confidence intervals.

The revised evaluation retains MAPE as a familiar percentage-based measure and adds weighted absolute percentage error (WAPE), symmetric MAPE (sMAPE), and mean forecast error (forecast minus actual). MAE and RMSE remain reported in millions of transactions per day. MAPE summarizes average absolute percentage error, WAPE weights errors by the total transaction volume, sMAPE scales the absolute error by the average magnitude of actual and predicted values, and mean error indicates whether forecasts tend to overpredict or underpredict. Reporting multiple metrics reduces reliance on any single score.

$$MAE = (1 / n) \times \Sigma \mid y(t) - \hat{y}(t) \mid \qquad RMSE = \sqrt{[\ (1 / n) \times \Sigma\ (\ y(t) - \hat{y}(t)\ )^2\ ]}$$

Errors were also split by day type: holiday-window days, sale days and ordinary days. This shows where each model gains or loses accuracy, which an overall average would hide.

5.6 Effect Recovery and Diagnostics 8

The full Prophet model was fitted on all the data, and its estimated effect for each holiday and event was compared with the true injected effect. Residuals were checked with the Ljung-Box test for autocorrelation [7] and the ARCH-LM test for changing variance [8]. All work was done in Python using prophet, statsmodels, pandas and matplotlib.

6.0 Results and Analysis

6.1 Exploratory Findings

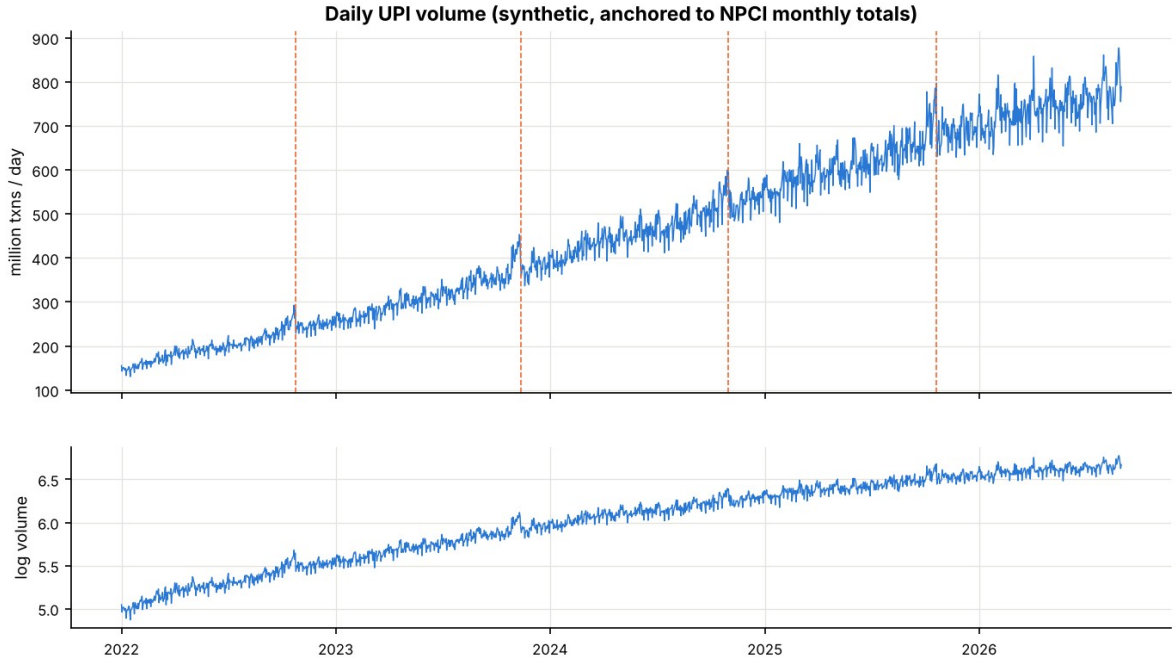


Figure 1: Daily UPI volume (top) and log volume (bottom); dashed lines mark Diwali

Volume grows about fivefold over the period, and the growth is close to linear on the log scale. The weekly up-and-down pattern grows with the level of the series, which supports multiplicative seasonality. Festival seasons appear as bumps each October and November.

Table 3: Stationarity tests

Series	ADF p-value	KPSS p-value	Result
log y	0.0739	0.01	Non-stationary
First difference of log y	0.0000	0.10	Stationary

Log volume is non-stationary, and one difference makes it stationary, so SARIMAX uses d = 1. The ACF of the differenced series (Figure 2) has strong spikes at lags 7, 14, 21 and 28, showing a weekly cycle, so the seasonal period is 7.

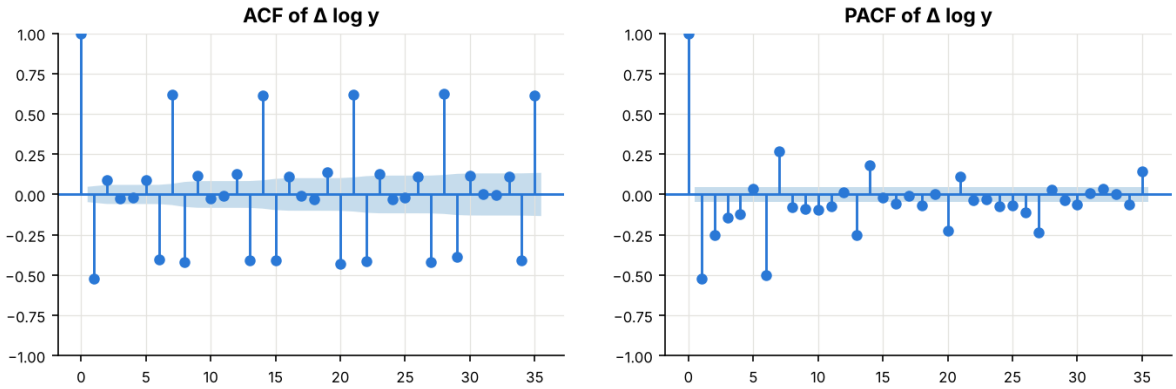


Figure 2: ACF and PACF of the differenced log series

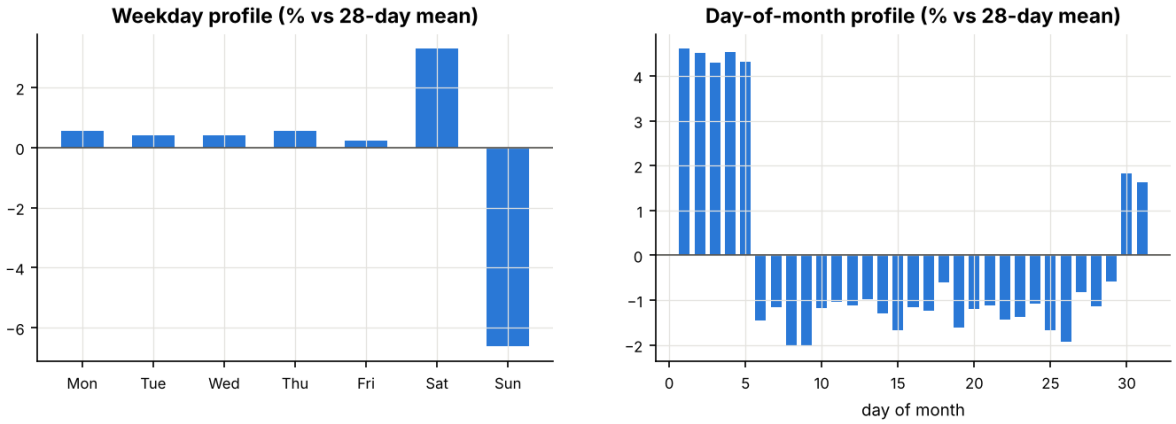


Figure 3: Average weekday and day-of-month profiles (ordinary days only)

Saturday volume is about 3% above the weekly average and Sunday about 6.5% below. Days 1 to 5 of each month are about 4.5% above the 28-day average, and the last two days about 1.7% above it. Both patterns are recurring calendar effects that the models must capture.

6.2 SARIMAX Order Selection

Table 4: Five best SARIMAX models by AIC (2022 to 2023)

(p, q, P, Q)	AIC	BIC
(1, 1, 1, 1)	-3091.9	-2807.2
(1, 2, 1, 1)	-3083.1	-2793.8
(2, 2, 1, 1)	-3081.0	-2787.1
(0, 2, 1, 1)	-3070.5	-2785.8
(2, 1, 1, 1)	-3069.4	-2780.1

The AIC search selected SARIMAX(1,1,1)(1,0,1) with a weekly period. All five best models include seasonal AR and MA terms, so the weekly pattern matters most. The residual diagnostics (Figure 4) look close to white noise: the Ljung-Box test finds no significant autocorrelation up to lag 7 ($p = 0.12$), and only a small amount by lag 14 ($p = 0.01$).

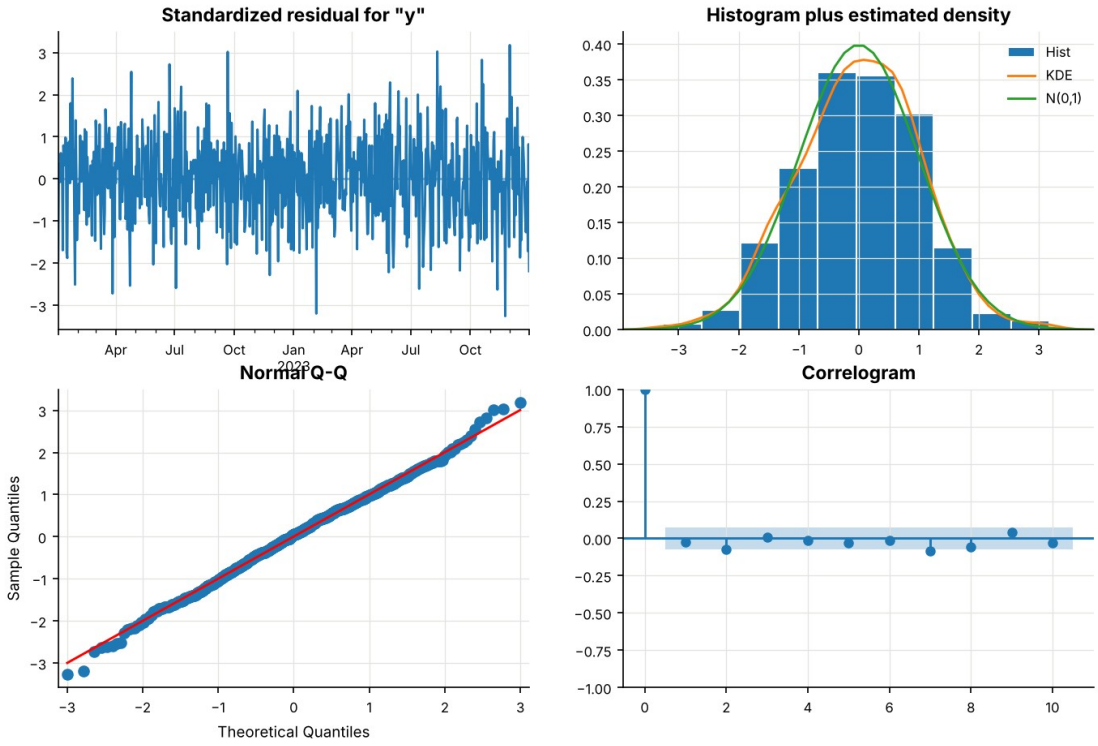


Figure 4: Residual diagnostics of the selected SARIMAX model

6.3 Forecast Accuracy

Table 5: Cross-validated forecast accuracy (10 folds, 90-day horizon)

Model	MAPE %	MAE	RMSE	Ordinary %	Holiday %	Sale %
Seasonal naive	5.54	32.48	41.08	5.29	7.88	11.01
SARIMAX	4.96	29.45	37.72	4.81	6.83	5.07
Prophet base	3.32	19.48	25.43	3.11	5.21	7.27
Prophet + holidays	3.17	18.58	23.92	3.07	3.63	8.57
Prophet + holidays + events	2.45	14.25	18.21	2.37	3.10	4.33

In the original notebook run, Prophet with holidays and events had the lowest reported MAPE (2.45%), compared with 5.54% for the seasonal-naive baseline and 4.96% for SARIMAX. Adding holidays reduced holiday-window MAPE from 5.21% to 3.63%, while adding events reduced overall MAPE from 3.17% to 2.45% and sale-day MAPE from 8.57% to 4.33%. These values are retained here as the original-run reference, not as verified outputs from the revised notebook. The revised notebook adds WAPE, sMAPE, mean error, horizon-specific results and fold-level variability, and applies a lognormal mean correction to SARIMAX. Therefore, all performance tables and statements about the winning model must be regenerated by running the revised notebook from top to bottom before final submission.

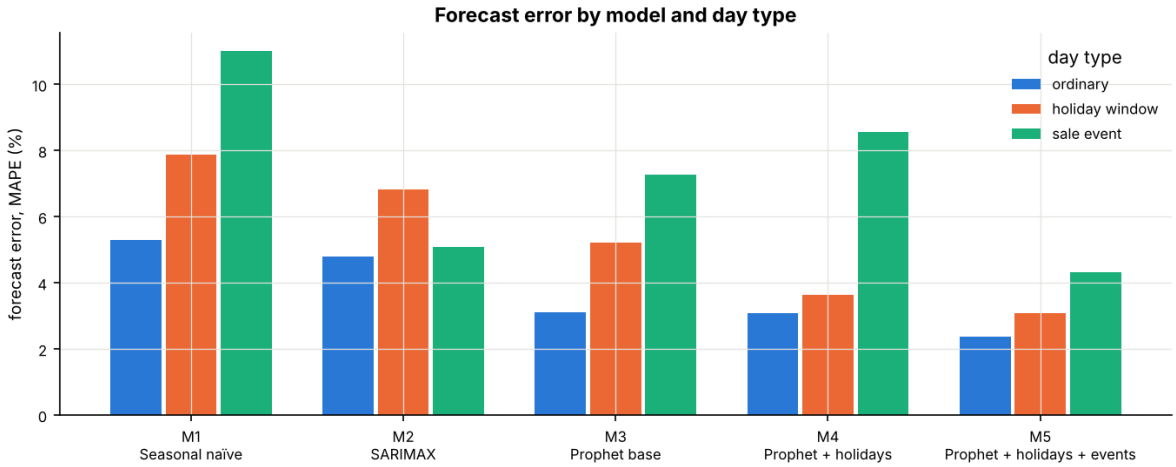


Figure 5: Forecast error by model and day type

SARIMAX was given the same calendar information as the full Prophet model. In the original run, its 90-day MAPE was 4.96%, and its errors were reported to grow across the forecast horizon. The revised evaluation now separates the first 7, 30 and 90 lead days and includes a lognormal mean correction for forecasts returned to the original scale. The horizon-specific tables should be used to determine whether the performance gap is consistent at short and long lead times; updated numerical conclusions should be written only after the revised notebook has been executed.

6.4 Holiday and Event Effects

Table 6: Estimated and true effects (mean over each window, %)

Effect	True	Estimated	Error (pp)
Diwali	9.14	7.75	-1.39
Holi	1.50	0.87	-0.63
Eid al-Fitr	4.50	3.91	-0.59
Raksha Bandhan	3.87	5.39	+1.51
Republic Day	-5.00	-6.86	-1.86
Independence Day	-5.00	-7.58	-2.58
Gandhi Jayanti	-5.00	-6.65	-1.65
E-commerce sale	10.67	9.80	-0.88
Payday (days 1 to 5)	6.00	6.17	+0.17
Month-end (last 2 days)	3.00	3.25	+0.25

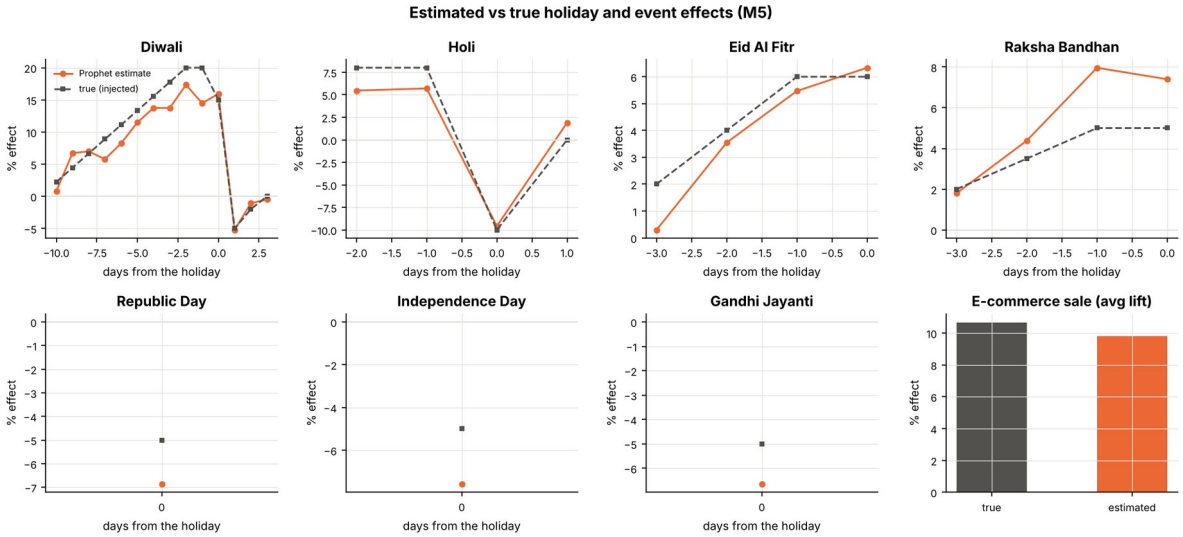


Figure 6: Estimated (orange) and true (dashed) effect for each day of each holiday window

Events were recovered almost exactly: payday within 0.2 percentage points and month-end within 0.3. The sale was estimated at 9.8% against a true 10.7%; its real lift varies between 8% and 12% from year to year, while the model fits one average. Diwali's shape was recovered, including the ten-day ramp, the peak on Dhanteras and Diwali eve and the dip after, but its size was underestimated: a peak of about 17% against 20%. Holi, Eid and Raksha Bandhan were within about 1.5 points, and Holi's -10% on the day itself was caught almost exactly. Single-day national holidays came out about 2 points too negative, as each occurs only four or five times.

6.5 Residual Diagnostics

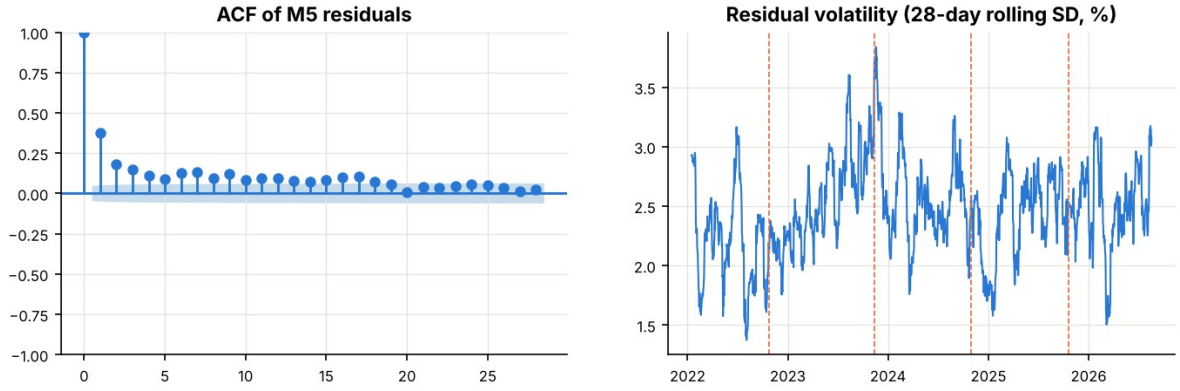


Figure 7: ACF of Prophet residuals (left) and residual volatility over time (right)

The residuals of the full Prophet model are autocorrelated: the lag-1 autocorrelation is 0.38, above the 0.3 built into the noise, and the Ljung-Box test rejects strongly ($p < 0.001$). Prophet assumes independent errors and does not model this. The ARCH-LM test ($LM = 31.5$, $p = 0.00005$) shows that the residual variance also changes over time, with the rolling standard deviation moving between about 1.5% and 3.8%. Prophet's prediction intervals assume constant variance, so they are too narrow in volatile periods.

6.6 Final Forecast

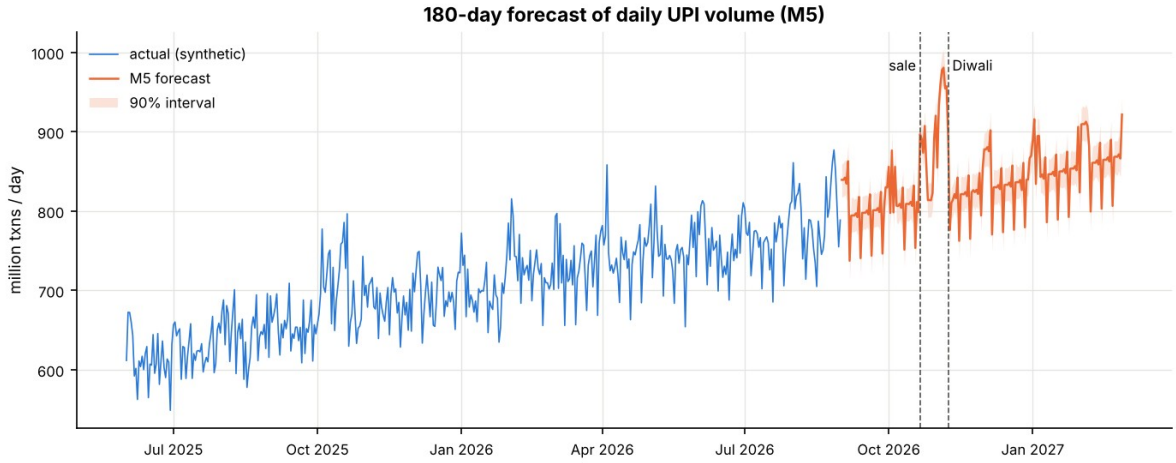


Figure 8: 180-day forecast with 90% interval (September 2026 to February 2027)

Table 7: Forecast monthly totals

Month	Forecast (billion transactions)	Days covered
September 2026	24.08	30
October 2026	25.74	31
November 2026	25.49	30
December 2026	25.99	31
January 2027	26.42	31
February 2027	23.51	27

The forecast continues the trend at about 24 to 26 billion transactions a month, against 24.5 billion in August 2026. February 2027 is lower only because the forecast ends on 27 February. The October 2026 sale and the Diwali ramp in early November give the highest forecast days, close to one billion transactions a day. These figures forecast the synthetic series and illustrate the method; they are not predictions of real UPI volume.

7.0 Discussion

The revised forecasting pipeline adds several evaluation safeguards. WAPE and sMAPE complement MAPE, while mean forecast error identifies systematic over- or underprediction. Results are reported for 7-, 30- and 90-day lead horizons and fold-level scores are summarized with their spread. These additions help distinguish near-term performance from longer-horizon performance. Since rolling-origin windows overlap, fold-level results are interpreted descriptively rather than as independent statistical replicates. The SARIMAX forecast is also adjusted using an approximate lognormal correction when transforming predictions from log volume back to transaction volume. The correction is a modelling assumption and must be judged by out-of-sample metrics.

7.1 Why Holidays and Events Help

Each improvement appeared on exactly the days the added information describes. Holidays improved holiday days, the sale improved sale days, and payday and month-end improved ordinary days. This shows that the gains come from the calendar knowledge itself, not from extra model flexibility.

7.2 Prophet Versus SARIMAX

SARIMAX had the cleaner in-sample residuals, yet it forecast worse than every Prophet version. AIC rewards a good fit to the training window, not accuracy 90 days ahead. ARIMA forecasts depend on recent values, so their errors build up with the horizon, while Prophet's forecast comes from a smooth trend and fixed calendar terms. For long-horizon planning of daily volume, Prophet is the better choice; for next-day forecasts, the gap would likely be smaller.

7.3 Why Diwali Is Underestimated

Diwali is the largest and longest effect. Each of its 14 daily coefficients is estimated from only four Diwalis in the data. Diwali also always falls in October or November, so the yearly seasonal pattern can absorb part of its effect, and the trend inherits some month-to-month variation from the real NPCI data around festive months. The earlier anchoring issue (Section 4.3) made this much worse; after the correction, about 1.4 points of the gap remain.

7.4 Limitations

- The daily data is synthetic, so the results show how the models behave under the specified data-generating assumptions, not the real size of festival effects or validated real-world UPI forecast accuracy. Five festive months differ from NPCI totals by design, and only four Diwalis are available for estimating the festival profile. Forecast folds overlap, so fold-level variation is descriptive rather than a formal independent-sample uncertainty estimate. The lognormal correction for SARIMAX is approximate and must be assessed on held-out forecasts. Prophet residuals were found to be autocorrelated and their variance changes over time; prediction-interval coverage should therefore be measured empirically on held-out data before intervals are used for operational planning.
- Five festive months differ from the NPCI totals by up to 6.5% by design.
- Only four Diwalis are in the sample.

- Prophet's residuals are autocorrelated and their variance changes, so its prediction intervals are too narrow.

8.0 Conclusion and Future Scope

This project implements Prophet to forecast daily UPI transaction volume with holiday and event effects, using a synthetic daily series anchored to real NPCI monthly totals. The original experiment reported that Prophet with holidays and events outperformed the seasonal-naive baseline and SARIMAX on the 90-day rolling-origin MAPE measure. The revised notebook strengthens the evaluation by adding WAPE, sMAPE and mean forecast error, examining 7-, 30- and 90-day forecast horizons, summarizing fold-level variability, validating data inputs and applying an approximate lognormal mean correction to SARIMAX forecasts. The updated numerical results must be regenerated before final submission; the original reported scores should not be presented as outputs of the modified pipeline. The simulation also permits comparison of estimated calendar effects with their known injected values, but this is a simulation check rather than a causal estimate of actual UPI behaviour. Prophet residuals were previously found to be autocorrelated and heteroskedastic, so prediction intervals require further out-of-sample coverage assessment before operational use.

Reporting note: The revised notebook now writes results_table.csv, horizon_results.csv, fold_metrics.csv and fold_summary.csv under the results/ directory. Insert the regenerated tables and figures into this report after executing the updated notebook. Do not copy the original MAPE values into the revised results tables without verification.

Future work could:

- use real daily payment data if NPCI or RBI publish it;\n ● compare models across 7-, 30- and 90-day lead horizons and retain fold-level error summaries;\n ● measure empirical coverage of the nominal 90% prediction intervals on held-out data;\n ● test whether the SARIMAX lognormal mean correction improves validation metrics;\n ● combine Prophet with an ARIMA model of its residuals, or a GARCH model of changing variance;
- combine Prophet with an ARIMA model of its residuals, or a GARCH model of their changing variance;
- detect unannounced events automatically from forecast errors;
- compare with machine learning models such as LSTM networks;
- model P2P and P2M volume together with VAR models to study how festivals shift the payment mix.

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